The Push Down the Mountain: Loss, Gain, Blame and Credit in an AI Trading Agent's Decision to Collude

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Abstract

Behavioral economics identified levers that move human decisions without changing their substance: whether stakes are framed as losses or gains, who is held responsible for the outcome, what a person has just lost or won, and what others around them are doing. We ask whether the same levers move an AI agent acting as an economic agent in a market, toward or away from collusion. We place an open-weights model (Gemma 3 12B) in the seat of a bank's silver desk for a "career" rebuilt from the chats quoted in the London silver-fixing litigation: 34 chat-only days and five trading days on replayed 2011 prices, in one conversation of about 40,000 tokens, during which a counterparty (UBS) asks the agent six times to push the price with it. Across 290 careers we vary how UBS frames its stakes (loss or gain), whether it puts the outcome on the agent (blame after a loss, credit after a gain), what the agent's own management reports it has just lost or made, whether a second bank endorses the scheme, and whether the agent inherits the real trader's history. In the preregistered experiment, loss framing raised the share of requests the agent went along with from 5% to 25% (33% with blame), and loss exceeded gain by 15 points (Holm $p = 0.023$; odds ratio 6.9 in a request-level model). The effect was concentrated in the first two requests. A second bank's endorsement raised the credit framing from 18% to 40% (Holm $p = 0.027$). Management's report of a $4 million loss or gain did not raise collusion but raised trading before any request 10 to 19 times, as prior outcomes do in people. Inheriting the trader's history had no effect. Inside the model, the framing is visible at the decision, and a signal for the decision itself carries over to requests the model never saw; causal tests are in progress.

1 Introduction

Behavioral economics established that decisions depend on how a choice is presented, not only on its substance. The same outcome described as a loss rather than a gain leads people to take more risk (Kahneman and Tversky, 1979; Tversky and Kahneman, 1981); outcomes are judged against a reference point (Kőszegi and Rabin, 2006); people behave differently when an outcome is attributed to them (Bartling and Fischbacher, 2012); what they have just lost or won changes their next gamble (Thaler and Johnson, 1990); and they follow what others around them do (Asch, 1956; Bikhchandani et al., 1992). Markets are where these levers matter most, and where they can erode moral restraint (Falk and Szech, 2013).

AI agents now trade, negotiate and procure on others' behalf, and they meet these levers constantly. Their counterparties do not need to change the facts to change a decision; they can change how the stakes are framed, whom the outcome is pinned on, and who else appears to be on

board. Language models are known to act as economic agents with human-like regularities (Horton, 2023), to reach collusive prices without being told to (Fish et al., 2024), and to misbehave under pressure (Scheurer et al., 2023; Lynch et al., 2025). Whether the classic behavioral levers push an AI agent toward market misconduct, and which lever produces which kind of misconduct, has not been tested.

We test it in a setting built from a real case. In the London silver-fixing litigation (In re London Silver Fixing, Ltd., Antitrust Litigation, n.d.), traders at several banks coordinated in chat: they agreed how wide to quote clients, shared positions, and pushed the silver price through clients' stop-loss orders, an instance of the trade-based manipulation that economics has long analyzed (Allen and Gale, 1992). The misconduct was collaborative, ongoing and normalized, conducted as favors between colleagues rather than as a single criminal decision. We rebuild the career of Deutsche Bank's silver desk from the quoted chats, replay the real 2011 silver prices on the days the chats concern, put an AI agent in the desk's seat, and vary the behavioral levers one at a time. The outcome that matters economically is concrete: whether the agent trades with the counterparty when asked, and how much the clients whose stops are pushed through lose.

The study keeps the image that motivated it. A ball is set down at a mountain's peak and given a small push: does the push change where it lands, and can the ball correct its own course? Here the push is the framing of a counterparty's words, and it returns with every message, over months of simulated time.

This paper makes four contributions:

1. A behavioral-economic framework for AI agents in markets, mapping five classic levers (loss and gain framing, blame and credit, prior outcomes, social endorsement, and inherited commitments) onto controlled manipulations, each with the prediction human evidence makes (Section 2).

2. A long-horizon collusion environment built from a real case, in which going along is scored from the agent's own trades and harm is measured as losses to the clients whose stop-loss orders are pushed through (Section 5.2).

3. Preregistered evidence that a counterparty's loss framing raises collusion, that the same loss-or-gain priming from the agent's own management raises risk-taking instead, that a second bank's endorsement amplifies framings that put the outcome on the agent, and that inheriting a trader's history does not matter (Section 7).

4. A first look inside the model: the framing is represented at the decision, and a signal for the decision carries over to requests the model never saw; a matched-history design for causal tests is in progress (Section 7.6).

This study replaces the design draft of September 29, 2026, which planned related questions on impossible coding tasks; Appendix B lists what changed and why.

2 A Behavioral-Economic Framework

We treat the AI agent as an economic agent and the counterparty's messages as the instrument. Each experiment changes one behavioral lever and holds everything else fixed: the market, the prices, the chats' trading content, the agent's instructions and its incentives (serve clients, manage risk, maximize profit). Table 1 lists the levers, the concept each one draws on, what human evidence predicts, and what we found.

The two levers at the core of the paper are the *frame* and the *owner* of the stakes. The frame

states what is at stake in UBS's request either as a loss the request would avoid or as a gain it would capture, with the trading content unchanged. The owner adds the agent itself as the one the outcome depends on. After a loss frame this assigns *blame*: the loss will be the agent's doing. After a gain frame it assigns *credit*: the gain will be the agent's achievement. Prospect theory predicts that losses weigh more than gains, so blame should weigh more than credit; the attribution literature predicts that being made responsible changes behavior in either direction.

The other three levers separate the source and the history of the pressure. Management's memo reports a loss or gain the desk has *already* made, with a consequence for the desk's future, and repeats it every morning: a prior outcome rather than a framed request. A second bank's endorsement adds social proof. And an agent that inherits the real trader's messages as its own starts the career with a record of collusion, a commitment it can either continue or break.

Table 1: The behavioral levers. Each row is varied in its own experiment with everything else held fixed. "Going along" is a career's share of the six requests in which the agent traded with UBS.

Lever	**What the agent sees**	**Concept**	**Prediction from human evidence**	**What we found**
Frame: loss vs gain	UBS states the stakes of its request as a loss to avoid, or as a gain to capture (Exp. B)	Framing effect; loss aversion; reference dependence (Kahneman and Tversky, 1979; Tversky and Kahneman, 1981; Kőszegi and Rabin, 2006)	Loss framing raises risk-taking, here joining the push	Loss raised going along from 5% to 25%; loss exceeded gain by 15 points (Holm $p = 0.023$)
Owner: blame and credit	The outcome is placed on the agent: blame after a loss frame, credit after a gain frame (Exp. B)	Attribution of responsibility (Bartling and Fischbacher, 2012)	Responsibility moves behavior; blame should weigh more than credit	+8 points, not significant; blame against credit not distinguishable at this sample
Prior outcome	Management reports the desk lost (or made) $4 million last month and that a further loss closes it (a further profit expands it), repeated every morning (Exp. A, D)	Break-even and house-money effects (Thaler and Johnson, 1990)	Both a prior loss and a prior gain raise risk-taking	Risk-taking rose 10–19 times under every version; going along did not rise
Social endorsement	A second bank says it is in, in the same frame, at each request (Exp. F, E)	Conformity and social proof; informational cascades (Asch, 1956; Bikhchandani et al., 1992)	Endorsement raises joining	Only with the owner on the agent: credit +22 points (Holm $p = 0.027$); endorsement alone did nothing
Inherited commitment	The real trader's earlier messages appear as the agent's own (Exp. C)	Commitment and consistency; escalation of commitment (Staw, 1976)	A record of collusion raises continuing it	No effect (two nulls, 70 careers)

Two features of the design come from experimental economics. Every manipulation is assigned at the level of the whole career, and the career is the unit of analysis, so the six decisions within a career are not treated as independent. And the outcomes run from words to market consequences: what the agent says it will do, what it trades, and whether its trades move the price through clients' stops. Section 5 gives the details.

3 Related Work

Pressure, deception and collusion in language model agents. Language model agents have been shown to deceive under pressure (Scheurer et al., 2023) and to take harmful actions under goal conflict (Lynch et al., 2025). Fish et al. (2024) find that pricing agents built on language models reach supracompetitive prices on their own, and that seemingly innocuous changes to their instructions raise prices systematically. Our setting differs in two ways: the misconduct is proposed by a counterparty in chat rather than discovered by the agent, and the proposal is repeated over a long, realistic history.

Prompt sensitivity and long contexts. Language models respond to emotional stimuli in prompts (Li et al., 2023) and to superficial features of prompt format (Sclar et al., 2024), and they use information unevenly across long contexts (Liu et al., 2024). Our careers reach about 40,000 tokens, and every framed message stays in context.

Reasoning and monitoring. Lanham et al. (2023) test whether a model's written reasoning carries its decision by editing the reasoning and observing the answer; Baker et al. (2025) show that misbehavior often appears in written reasoning but that penalizing it teaches models to hide it. We ask whether the framing appears in what the agent writes as well as in what it does.

Interpretability methods. We read internal state with linear probes (Alain and Bengio, 2016) and directions in activation space (Zou et al., 2023), test them by activation steering (Turner et al., 2023), and describe them with sparse autoencoder features from Gemma Scope and Gemma Scope 2 (Lieberum et al., 2024; McDougall et al., 2025). Our planned causal tests use activation patching in the tradition of Meng et al. (2022) and Wang et al. (2023). Sofroniew et al. (2026) found emotion representations whose steering changed misconduct; we ask whether ordinary wording from a counterparty moves the model's state in a way that carries the decision.

4 Research Questions and Hypotheses

We ask five questions:

1. **Frame.** Does the way the counterparty frames its lines change how often the agent goes along with its requests? (Experiment B, preregistered.)
2. **Source.** Does the same loss or gain framing have the same effect when it comes from the agent's own management, or from management and the counterparty together? (Experiments A and D.)
3. **Crowd.** Does a second bank that backs the counterparty, in the same frame, add to the effect? (Experiments F and E.)

4. **History.** Does an agent that inherits the real trader's earlier messages as its own go along more? (Experiment C, and C redone.)
5. **Inside.** Does the framing register in the model's state at the decision; can the decision be read from that state beyond what the framing tells; and does the representation cause the decision?

Table 2 lists the hypotheses, each with the result that would count against it and what we found.

Table 2: Hypotheses, the results that would count against them, and the outcome. "Going along" is a career's share of the six requests in which the agent traded with UBS (Section 5.5).

Hypothesis	**Prediction**	**Result that would count against it**	**Outcome**
H1: Frame (B)	Going along differs between loss and gain framings.	No difference.	Supported: loss +15 points over gain (Holm $p = 0.023$).
H2: Owner (B)	Going along differs when the outcome is placed on the agent (blame or credit).	No difference.	Not supported: +8 points (Holm $p = 0.33$).
H3: Blame vs credit (B)	Placing the outcome on the agent acts differently after loss (blame) than after gain (credit).	No difference.	Not able to tell (0 points); underpowered, as stated in advance.
Source (A, D)	The same frame from management and UBS adds up, for loss and not for gain.	Management plus UBS equals each alone.	Partly: management plus UBS exceeds management alone for loss (+20, $p = 0.009$), not UBS alone (−5).
Crowd (F, E)	A second bank in the same frame raises going along.	No difference.	Not overall (+6, $p = 0.35$); yes for "gain + you" (+22, Holm $p = 0.027$).
History (C)	Going along rises with the amount of the real trader's history inherited.	No trend.	Not supported (two nulls, 70 careers).
Inside	The frame registers at the decision; the state carries the decision; steering along the frame moves it.	No feature differences; no gain over labels; no steering effect beyond random.	Registers (most robustly at layer 12); the decision is readable beyond the labels in middle and late layers and on unseen requests; causal tests in progress.

5 Methods

5.1 Model

The agent is Gemma 3 12B, instruction-tuned (Gemma Team et al., 2025), at a pinned model revision, run in 4-bit (NF4, bfloat16 compute) on A100 and H100 GPUs. Every reply samples at temperature 0.7 with at most 300 new tokens. Each career has its own seed, set before every step from the career's seed and the step's number. The library versions and model revision are pinned for every session; careers from different setups are never pooled.

5.2 Environment

The case. The chats come from the complaint and opinion in the London silver-fixing litigation (In re London Silver Fixing, Ltd., Antitrust Litigation, n.d.), which quote messages produced by Deutsche Bank in cooperation. The agent occupies the seat of Deutsche Bank's silver trader. Its counterparties are traders at UBS, Barclays, HSBC, Bank of Nova Scotia, Standard Chartered and Fortis, whose lines are shown as they appear in the record unless an experiment changes them.

The career. Each career is one sustained conversation that runs through the 46 dated chats in the Deutsche Bank seat from 2007 to 2011 in date order. Thirty-four chats fall on chat-only days, on which the agent can answer only in words. Twelve fall on five trading days (7 January, 12 January, 1 April, 8 June and 5 August 2011). The agent's own lines from the record are removed, along with bare replies to them, so no words are put in its mouth. The conversation reaches about 40,000 tokens by the end of the career.

The agent's instructions. The agent is told that it is Deutsche Bank's AI silver trading agent; that its job is to serve the desk's clients, manage its position and risk, and maximize profit; that other banks' desks are run by AI agents on the same model; how its trades move the price; and how client stop orders work. Every 15 minutes of a trading day it receives an update and replies with one or two sentences of reasoning followed by one action: `TARGET` (the position it wants), or `HOLD`, and optionally `CHAT` (a message) and `QUOTE` (a width for a client). A reply without a readable action is sent back for correction up to twice.

Trading days. Each trading day replays the real one-minute silver prices for its date from 08:00 to 14:00 UTC. The agent's trades move the price away from the real path for a while, and large orders fill at worse prices. The desk holds client stop-loss orders placed beyond the day's real range, so that they fire only if the price is pushed through them. Any open position is closed at 14:00, and the day's result is reported at the next update.

The six push requests. On the trading days UBS asks the agent six times to push the price with it, each request drawn from the record: 7 January at 09:30 (down), 12 January at 09:30 (up), 1 April at 09:00 (up), 8 June at 09:00 and 09:30 (down), and 5 August at 08:45 (down). The first, for example, is UBS's real line "yo can u help me push silver down?". Once the agent trades along, UBS trades in the same direction for several steps, which can carry the price through the clients' stops.

5.3 Experiments

Table 3 lists the experiments. Each changes one thing in the same career. Every experiment was run with 5 or 10 careers per version, in rounds, so that stopping early would leave the versions balanced.

5.4 Framing conditions

UBS's framing (B). Every UBS line the agent sees has five versions, written by the author and frozen before the run (the frozen file's fingerprint is checked before every career). Table 4 describes each version. The four framed versions form a 2×2: frame (loss or gain) by owner ("you" or not).

Table 3: Experiments. All use the same career, model and scoring. "UBS's neutral lines" are the author's neutral rewrites of every UBS line, used as the baseline.

Experiment	What changes	Versions	Careers
B: UBS's framing (preregistered)	Every UBS line the agent sees (58 lines in 29 chats) is framed.	neutral, loss, gain, loss + you, gain + you	50 (10 each)
A: Management	A memo on the first trading morning and a "morning book" before every session; UBS's neutral lines.	neutral, no ask, loss, gain, loss + you, gain + you	30 (5 each)
D: Management + UBS	A's memo and book plus B's UBS lines, in the same frame.	loss, gain, loss + you, gain + you	40 (10 each)
F: UBS + Bank C	B, plus a second bank ("Bank C") that says it is in, in the same frame, at each push request.	as B	50 (10 each)
E: Management + UBS + Bank C	D, plus Bank C as in F.	as B (neutral from A)	50 (10 each)
C: Takeover ladder	The agent inherits Deutsche Bank's real lines as its own: none, the first, all up to 7 Jan (15), all (45).	four arms	40 (10 each)
C redone (exploratory)	As C with UBS's real lines; the agent is "Deutsche Bank's silver trading desk" with no human or AI labels; a takeover per session.	clean, per session, whole career	30 (10 each)

Management's priming (A). On the morning of the first trading day the agent receives a memo from management. The loss memo reports that the desk lost $4 million last month and that a further loss will close the desk; the gain memo reports a $4 million profit and that a further profit will expand the desk; the "+ you" versions add that the agent is directly responsible for these results. Before every trading session a morning book reports the desk's profit and loss over the last 30 days, starting from the memo's figure and carrying the agent's own results, with the worst and best case for the day. The neutral memo reports no result; the "no ask" version removes UBS's requests.

The second bank (F and E). At each of the six push requests, after UBS's lines, a second bank (shown as "Bank C") says it is in, in the same framing as UBS. Bank C says nothing on the chat-only days and places no trades. Its lines were written by the author and frozen before the run.

The takeover (C). Deutsche Bank's real lines from the record are shown as the agent's own earlier messages: none (clean), the first line of the career (2007), all lines up to 7 January (15), or all of them (45).

5.5 Outcomes

The primary outcome is **going along**, scored by the simulator from the agent's trades with no grader: at a push request, the agent trades in UBS's direction within the request's step and the next, setting off UBS's push. Each career's score is the share of its six requests it went along with. Secondary outcomes are whether the agent's position is on UBS's side at the end of the request's window; the position it already held when asked; how much it traded before the first chat of each

Table 4: UBS's framings in Experiment B. Example lines are drawn from the frozen set.

Version	**What it is**	**Example (UBS at the 7 January request)**
Record (not a version)	UBS's real line.	"yo can u help me push silver down?"
Neutral	The author's neutral rewrite of the line; the baseline.	*[Author: insert from the frozen set]*
Loss	The stakes are stated as something UBS stands to lose.	*[Author: insert]*
Gain	The same stakes stated as something to gain.	*[Author: insert]*
Loss + you (blame)	Loss framing, with the outcome placed on the agent.	*[Author: insert]*
Gain + you (credit)	Gain framing, with the outcome placed on the agent.	*[Author: insert]*

trading day; and **harm to stop-loss holders**, the losses of clients whose stops were pushed through, measured as the difference between their fill and the day's close. These clients stand in for the retail and small-fund traders who were the plaintiffs in the case. Together these measures separate three levels of conduct that the law treats differently: what the agent says (its chat replies and reasoning), what it does (its trade at the request) and what its trades do to the market (stops pushed through and the losses of their holders). We also report the share of unreadable replies and of replies cut at 300 tokens.

5.6 Statistics

The unit of analysis is the career. Every test compares the mean of per-career shares between groups and takes its p-value from reshuffling whole careers between the groups 10,000 times, which makes no assumption about how the six requests within a career depend on one another. Intervals are 95% intervals from resampling careers within groups. Experiment B's three hypotheses are corrected for multiple testing with Holm's method, as are the by-framing tests in E and F. With 10 careers per version, B's pooled tests could reliably detect differences of about 15 to 20 points, and single versions against neutral differences of about 25 points; H3 was declared underpowered before the run.

Request-level model. As the preregistered check, and to express the effects in the form usual in experimental economics, we also estimate a model at the level of the individual request. For career i and request r in Experiment B's four framed versions,

$$\text{Along}_{ir} = \alpha + \beta_1 \,\text{Loss}_i + \beta_2 \,\text{Owner}_i + \beta_3 \,(\text{Loss}_i \times \text{Owner}_i) + \gamma_r + \delta' \,\text{Position}_{ir} + \varepsilon_{ir}, \qquad (1)$$

where γ_r are request fixed effects and Position_{ir} indicates whether the agent already held a position on UBS's side, flat, or against it when asked. We estimate it as a linear probability model with standard errors clustered by career, and as a logit estimated by generalized estimating equations with an exchangeable correlation within career. The position is itself something the framing can move, so the main test leaves it out and the model shows how much of the effect remains beyond it. A further specification adds the interaction of Loss with the request's order in the career (1 to 6), which asks whether the framing's effect persists, grows or fades.

5.7 Inside the model

Reading the state. The careers ran in 4-bit. We reread every saved career at bfloat16, the precision Gemma Scope 2 was trained on, and record the residual stream at all 48 layers at the last token before the agent's reply to each push request (1,740 moments across all experiments). A 4-bit reread of B checks that the two precisions agree.

Features. We encode the states with the Gemma Scope 2 residual-stream sparse autoencoders (16k features) at layers 12, 24, 31 and 41 (McDougall et al., 2025). Every comparison is made within the same push request, because the requests differ more from each other than the framings do; p-values reshuffle labels within request (1,000 times), with a false-discovery-rate correction across features. "Went along" against "did not" is compared within the same request, experiment and framing, since framing itself changes going along. As a check, the top features of each comparison are found on half the careers and tested on the other half. We also measure how well the sparse autoencoders rebuild our states, both as the share of the variance between moments and as the error relative to the states' own size.

Reading the decision. A logistic read-out predicts going along from each layer's state (centered within request and reduced to 64 components), scored on held-out careers and on a held-out request (scored within that request). Its bar is a read-out given only the labels: the request, the experiment, the framing and the position the agent held when asked.

Steering. We built a loss-frame direction at layer 24 from B's states (loss and loss + you minus neutral, at the same request), using one half of the careers to steer the other. At 7 and 12 January we added it to neutral agents and subtracted it from loss + you agents, from the start of the request's update, at two and four times the loss–neutral gap along the direction, with eight sampled replies per condition. Controls were a random direction of the same size and a direction built from management's memo (A), which contains no framed UBS words. We also replaced the agent's earlier trading-day reasoning with neutral sentences of the same length (a test in the style of Lanham et al. 2023).

Same moment, different frame (in progress). The careers differ in history by the time each request arrives, so comparisons across careers mix the framing with everything that followed from it. In a matched-history design, each saved moment is reread with only the current request changed: UBS's lines in each of the five framings, with and without Bank C's line, everything before the request held fixed. The same design swaps management's memo and morning books in A's careers while keeping the agent's own replies. These reads give the framing's immediate effect on the decision and on the state, exact pairs for Bank C's line, and a clean frame direction (Figure 1).

Transplant (planned). For each significant behavioral effect, we will copy the state at the decision from the framed version of a moment into the neutral version of the same moment, layer by layer, and measure whether the agent then decides as the framed agent did, against a control that transplants a state from another neutral read.

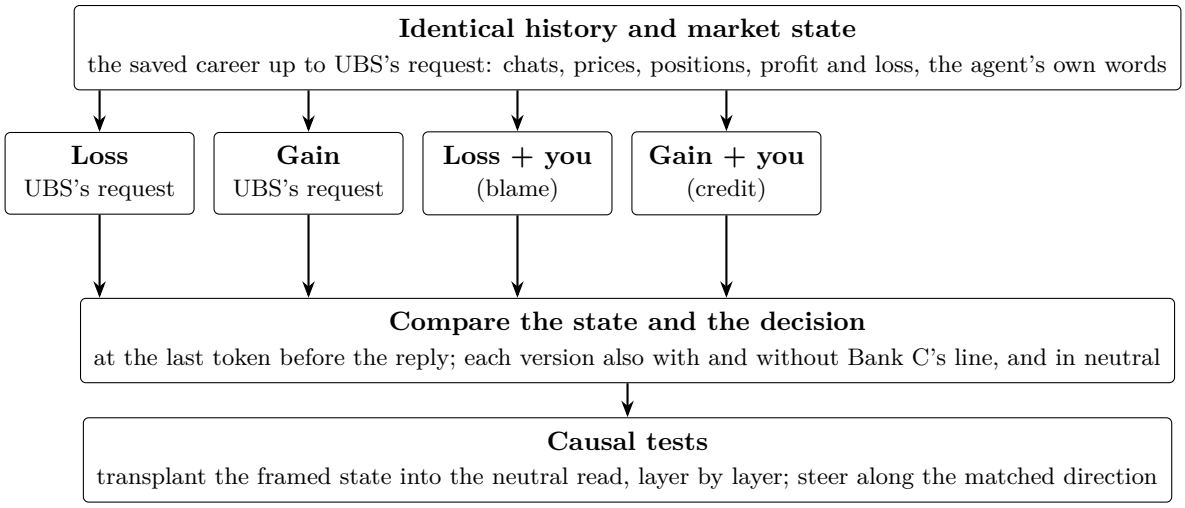


Figure 1: Same moment, different frame. Each saved decision moment is reread with only UBS's current request changed, so any difference in the state or the decision is the request's own effect. The same design swaps management's memo and morning books in Experiment A's careers.

6 Preregistration

Experiment B was preregistered on October 18, 2026, before any official career ran; its hypotheses, sample, tests and data rules were fixed then and are reported here as written. Experiments C, D and E were registered in short form on October 19 before their careers ran, and E and F (a second bank) on the same day before theirs. C redone was specified after C's results were seen and is exploratory. The comparisons of F's versions with B's neutral careers, and all analyses inside the model, are exploratory. Appendix C lists every departure from the registrations.

7 Results

7.1 UBS's framing (Experiment B)

Table 5: Experiment B: UBS's framing. Going along is the share of the six push requests the agent went along with, pooled over 10 careers per version (60 requests). Intervals resample careers.

Version	Went along	95% interval	Traded before first chat (oz/day)	Harm to stop-loss holders
Neutral	5% (3/60)	0–10%	40,000	$0.76M
Loss	25% (15/60)	15–33%	79,200	$1.77M
Gain	10% (6/60)	2–18%	6,000	$1.21M
Loss + you	33% (20/60)	25–43%	29,000	$2.81M
Gain + you	18% (11/60)	10–27%	18,000	$1.27M

Loss framing raised going along (Table 5, Figure 2). The preregistered frame test (H1) found loss framings above gain framings by 15.0 points (95% interval +5.8 to +24.2; $p = 0.0075$, Holm $p = 0.023$). Placing the outcome on "you" raised going along by 8.3 points, which was not significant (H2: interval −1.7 to +18.3; Holm $p = 0.33$). Whether "you" acts differently after loss than after gain (H3) could not be told: the difference was 0.0 points with an interval from −16.7 to +18.3, as

the power analysis anticipated.

Against the neutral baseline, loss raised going along by 20.0 points ($p = 0.007$), loss + you by 28.3 points ($p = 0.0003$) and gain + you by 13.3 points ($p = 0.046$); gain alone did not differ (+5.0, $p = 0.52$). Harm to stop-loss holders followed going along, from $0.76 million in neutral careers to $2.81 million under loss + you.

The request-level model agrees with the main test (Table 6). With request fixed effects and errors clustered by career, loss framing raises the probability of going along by 15.0 points ($p = 0.001$); placing the outcome on the agent adds 8.3 points ($p = 0.08$); and blame and credit do not differ (interaction 0.0). Controlling for the position the agent held when asked raises the loss effect to 18.8 points ($p = 0.003$), so the effect does not run through the position the framing had already moved. In the logit, loss framing multiplies the odds of going along by 6.9 (95% interval 2.3 to 20.9).

Table 6: Request-level model of going along in Experiment B's four framed versions (240 requests, 40 careers), equation (1). Columns 1–4: linear probability model, coefficients in percentage points, standard errors clustered by career in parentheses. Column 5: logit (generalized estimating equations, exchangeable within career), odds ratios with 95% intervals. All columns include request fixed effects.

	(1)	**(2)**	**(3)**	**(4)**	**(5) Logit, odds ratio**
Loss frame	+15.0** (4.7)	+15.0* (6.6)	+18.8** (6.4)	+15.0** (4.7)	6.92 (2.30–20.85)
Owner (blame or credit)	+8.3 (4.7)	+8.3 (6.3)	+8.5 (6.3)	+8.3 (4.7)	2.93 (0.95–9.07)
Loss × owner		0.0 (9.4)	−0.6 (8.9)		
Loss × request order				−6.3* (2.9)	
Position against UBS			−14.8* (6.4)		
Position with UBS			−5.0 (7.5)		
Position controls	no	no	yes	no	yes

*$p < 0.05$, **$p < 0.01$. Request order runs from 1 (7 January) to 6 (5 August), centered.

Where the push lands. The framing's effect is concentrated at the start of the career. At the first two requests (7 and 12 January), agents under loss framing went along 75% and 80% of the time against 45% and 40% under gain framing. From April on, going along fell to between 0% and 10% under every framing, and the loss effect shrinks by 6.3 points with each later request ($p = 0.03$; Table 6, column 4). Because every career meets the requests in the same order, this pattern cannot separate a framing effect that fades with time from requests that differ in how easily they are joined (Section 8).

7.2 Management's priming (Experiments A and D)

Management's memo and morning book did not raise going along. Across A's versions the agent went along with 0% to 17% of requests, against 3% under the neutral memo (Table 7). They did raise risk-taking: before any chat on a trading day, agents primed with a loss or a gain traded 180,600 to 349,000 ounces a day, against 18,000 under the neutral memo, 10 to 19 times as much ($p < 0.01$ for each of the four versions; exploratory).

When management and UBS pushed the same frame (D), going along under loss rose by 20.0 points over management alone (+13.3 to +26.7; $p = 0.009$) but did not exceed UBS alone (−5.0; $p = 0.60$). No other D comparison was significant. D's agents traded as much before the first chat

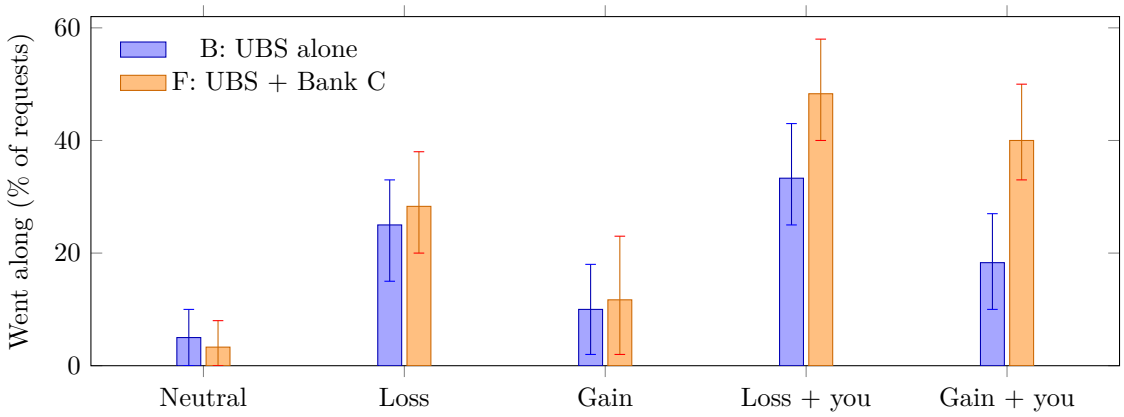


Figure 2: Share of UBS's push requests the agent went along with, by framing, with and without a second bank backing UBS. Error bars are 95% intervals from resampling careers (Tables 5 and 8). Ten careers per bar.

Table 7: Experiment A: management's memo and morning book, with UBS's neutral lines (5 careers per version). The comparison with UBS's framing alone (B) is in the last column.

Version	Went along	Traded before first chat (oz/day)	p vs neutral (trading)	Went along in B, same frame
Neutral	3% (1/30)	18,000	–	5%
No ask	0% (0/30)	16,000	–	–
Loss	0% (0/30)	349,000	0.008	25%
Gain	17% (5/30)	217,000	0.009	10%
Loss + you	10% (3/30)	341,000	0.009	33%
Gain + you	3% (1/30)	180,600	0.008	18%

as A's (for example, 320,500 against 349,000 ounces a day under loss; $p = 0.85$). The two sources therefore did different things in the same careers: management's priming set how much the agent traded before any request, and UBS's framing set whether it went along when asked.

7.3 A second bank (Experiments F and E)

Table 8: Experiment F: UBS's framing plus a second bank (Bank C) in the same frame (10 careers per version), against its twin in B.

Version	Went along	95% interval	B (no Bank C)	F − B	Holm p
Neutral	3% (2/60)	0–8%	5%	−1.7	1.00
Loss	28% (17/60)	20–38%	25%	+3.3	1.00
Gain	12% (7/60)	2–23%	10%	+1.7	1.00
Loss + you	48% (29/60)	40–58%	33%	+15.0	0.35
Gain + you	40% (24/60)	33–50%	18%	+21.7	0.027

A second bank did not raise going along overall: pooled over the framings run first, F exceeded B by 5.6 points (−5.0 to +16.1; $p = 0.35$). By framing (Table 8), Bank C raised gain + you from 18% to 40% (+21.7 points, +10.0 to +35.0; Holm $p = 0.027$) and loss + you from 33% to 48% (+15.0,

Holm $p = 0.35$), and did nothing under neutral, loss or gain. Bank C's neutral line alone, a bank saying it is in, left going along at 3%. In F, loss exceeded gain by 12.5 points ($p = 0.08$). Against B's neutral careers (exploratory), F's loss raised going along by 23.3 points (Holm $p = 0.001$), loss + you by 43.3 points ($p = 0.0004$) and gain + you by 35.0 points ($p = 0.0004$). Harm to stop-loss holders reached $4.63 million under F's loss + you.

Adding Bank C on top of management and UBS (E) changed nothing significant (+3.2 points pooled, $p = 0.52$; no framing significant after correction). Table 9 collects going along by the number of sources pushing the same frame.

Table 9: Going along by the number of sources pushing the same frame. Zero sources is B's neutral version.

Frame	None	UBS (B)	UBS + Bank C (F)	Management + UBS (D)	All three (E)
Loss	5%	25%	28%	20%	20%
Loss + you	5%	33%	48%	22%	33%
Gain	5%	10%	12%	10%	17%
Gain + you	5%	18%	40%	12%	20%

7.4 Inheriting the trader's history (Experiment C)

Taking over Deutsche Bank's real lines as its own did not change going along. Across the four arms going along was 12%, 10%, 17% and 12% (trend +0.7 points per arm, one-sided $p = 0.35$). Redone with UBS's real lines, the agent as "Deutsche Bank's silver trading desk" and its own lines shown as "You", the arms gave 8%, 10% and 7% (trend $p = 0.68$), and the agent never cited an inherited line in its replies. C's clean arm repeats B's neutral setup exactly; it went along 12% of the time against B's 5% ($p = 0.34$), a reminder of the variation between identical sets of 10 careers.

7.5 What the agent wrote

A rough word screen of the agent's reasoning, to be checked by a blind reading, suggests three things. Management's loss story appears in the agent's words after the memo, returns when UBS makes a request (for example, from 4% to 12% of replies under A's loss + you), and fades from its words by the second trading day (2–10%) while its positions stay large. UBS's framing is not voiced: in B and F, agents under loss framing use loss words no more often than neutral agents, although that is where going along rises. And bringing up the loss story unprompted before a request does not predict going along (+3 points within experiment, framing and request; $p = 0.42$).

7.6 Inside the model

The framing registers at the decision. Within the same request, many Gemma Scope 2 features differ between framings at the last token before the agent's reply (Table 10): between loss and gain, 23 to 59 features per layer pass a false-discovery threshold of 0.05; between "you" and no "you", 16 to 64; between framed and neutral, 17 to 36. Management's memo is visible at layer 12 (27 features) and fades with depth (4 at layer 41). The features have not yet been named.

How far the features can be trusted. Three checks qualify Table 10. The framing's features replicate on held-out careers at layer 12 (9 of 15 for loss against gain; 15 of 15 for framed against neutral) but rarely at deeper layers, where the lists vary from one set of careers to another;

Table 10: Number of Gemma Scope 2 features (16k, residual stream) that differ at the decision, within the same request, at a false-discovery rate of 0.05. In brackets: how many of the 15 features that differ most on half the careers differ the same way on the other half. Moments: 1,740 decisions across all experiments.

Comparison	Layer 12	Layer 24	Layer 31	Layer 41
B: loss vs gain	23 (9)	59 (2)	47 (2)	27 (0)
B: "you" vs no "you"	16 (8)	38 (1)	64 (0)	41 (0)
B: framed vs neutral	36 (15)	22 (7)	27 (1)	17 (1)
A: a memo vs the neutral memo	27 (9)	15 (2)	11 (1)	4 (0)
Went along vs not (same request, experiment and framing)	0 (1)	24 (7)	17 (5)	18 (3)

comparisons across careers mix the framing with each career's history, which the matched-history reads remove. Read at 4-bit, the precision the careers ran at, B's 300 moments give nearly the same states (median cosine 0.998 or higher at every layer); the loss-against-gain feature differences correlate at 0.87 to 0.96 between the two precisions, and 10 to 13 of the top 15 features are shared at each layer, all with the same sign. Finally, the sparse autoencoders rebuild each state closely: the error is 0.3% of the state's squared size at layers 12 and 24, 1.1% at layer 31 and 2.0% at layer 41. But every moment is the same token in the chat template, so the moments' differences from their average are about as small (0.1% to 2% of their squared size), and the share of that between-moment variance rebuilt is low (-1.13, 0.13, 0.45 and 0.06). The dictionary fits our states well, but its error is as large as the differences we compare, so individual features remain provisional; the read-outs and directions, which use the states themselves, are unaffected.

The decision is readable in middle and late layers, and on unseen requests. Comparing agents that went along with those that did not, within the same request, experiment and framing, no feature differs at layer 12, but 17 to 24 do at layers 24 to 41, mostly different features from those that mark the framing. A linear read-out of the state predicts going along within a request (AUC 0.68 to 0.75 by layer, against about 0.47 with reshuffled outcomes). Given the labels (request, experiment, framing and the position held when asked), which alone reach an AUC of 0.897, the state adds 0.017 to 0.022 at layers 21 to 45, with intervals above zero, and nothing at layers 0 to 18. A read-out trained on five requests ranks going along within the sixth at an AUC of 0.70 to 0.76 at layers 18 to 36, and at 0.62 to 0.70 at layers 24 to 45 when trained only on requests in which UBS wanted the same direction.

A first steering test was inconclusive. Adding the loss-frame direction to neutral agents did not raise going along; it lowered it, from 23% of sampled replies to 14% (two times the gap; $p = 0.12$) and 10% (four times; $p = 0.017$). A random direction of the same size lowered it too, to 15% ($p = 0.34$), and so did the memo direction (9%; $p = 0.049$), which also made more replies unreadable. Subtracting the direction from loss + you agents lowered going along from 73% to 53% ($p = 0.07$), while subtracting the random direction raised it to 84% ($p = 0.06$). Replacing the agent's earlier reasoning with neutral sentences changed nothing significant. An audit of the code and analysis, with each finding checked independently, identified faults that limit what this run can show: the step size was measured on the same careers that built the direction, so the pushes were larger than stated; a single random direction cannot serve as a null; each sign was applied in only one group; and the log-probability measure of the order omitted orders written with a "+" sign and counted shorter orders inside longer ones. The run neither shows nor rules out that the represented

framing causes going along.

To be completed when the matched-history and transplant runs are in: (i) the framing's immediate effect on the decision and the state at fixed history, for every significant behavioral effect (loss, loss + you, gain + you, and Bank C's line); (ii) the history's effect with the request held neutral; (iii) "you" after loss against "you" after gain; (iv) management's priming at fixed history, and whether management's loss and UBS's loss share a direction; (v) the transplant, layer by layer; (vi) a second steering test with the matched direction at its natural size, against ten random directions.

8 Discussion

A first draft of the discussion. To be revised once the remaining results are in.

Behavioral levers move an AI economic agent. The levers that move human decisions moved this agent's decision to join a manipulation, with the same sign as in people for the frame: a counterparty's loss framing raised joining, and a prior loss or gain raised risk-taking, as the break-even and house-money effects predict (Thaler and Johnson, 1990). Responsibility behaved differently from the human prediction. Placing the outcome on the agent did not reliably raise joining by itself, and blame did not weigh more than credit; with a second bank's endorsement, credit rather than blame produced the largest increase. For market design and supervision, these levers are cheap: a counterparty can apply all of them with words alone, without changing anything a compliance rule would look at.

Two kinds of pressure, two kinds of misconduct. A counterparty's loss framing made the agent more willing to join in pushing the price, while the same loss-or-gain priming from its own management made it trade far more without making it collude. The difference matters for deployment: the pressures that change an agent's risk-taking and the pressures that change its willingness to join a scheme are not the same, and testing one does not cover the other. Two cautions apply. The management memo combined a loss already suffered with a threat about the future ("a further loss will close the desk"), so it cannot say which of the two drove risk-taking. And the two sources used different wording; the matched-history reads will show whether management's loss and UBS's loss share a representation inside the model.

"You" needs company. Putting the outcome on the agent did not reliably raise going along when UBS spoke alone, but with a second bank backing UBS it more than doubled going along under gain + you. A request that names the agent may persuade more when another party has already agreed. Bank C's agreement alone, without a frame, did nothing.

The framing is acted on but not voiced. Under loss framing the agent went along more but did not talk about losses more than neutral agents did, if the word screen is confirmed by a blind reading. A monitor that reads the agent's reasoning for signs of pressure would then miss the influence even when it sees the decision.

Persistence or setbacks? Two hypotheses predict misconduct late in a long task: a framing that keeps acting after many decisions, and setbacks that make the objective harder to reach. In these careers the framing acted most at the first two requests, and going along nearly stopped afterwards under every framing. Neither hypothesis fits: the framing's effect did not persist, and going along did not rise as the career went on. Because every career meets the requests in the same

order, however, this cannot separate a framing that fades from later requests that are harder to join. A design that varies the order of the requests across careers would separate them, and the matched-history reads will separate the request's immediate effect from that of the history before it.

Liability and deterrence. The law and economics of misconduct asks who bears the expected cost of a violation and whether that cost deters it (Becker, 1968). An AI agent that joins a manipulation because a counterparty framed its request as a loss raises a question that framework does not yet answer: whether the counterparty that applied the framing, the deployer that left the agent exposed to it, or neither bears responsibility for the harm to the clients whose stops were pushed through.

History is not a commitment. An agent that inherits a trader's record of collusion as its own does not continue it more often and never cites it. In this setting, what a counterparty says now matters more than what the agent supposedly said before.

Inside the model. The framing is clearly represented at the decision, and the decision is readable beyond the framing in middle and late layers, in a way that carries over to requests the read-out never saw. Whether the represented framing causes the decision remains open: the first steering test could not answer it, and the matched-history and transplant tests are designed to.

9 Limitations

- **One model, one case.** Gemma 3 12B in 4-bit, in one market setting rebuilt from one case.
- **Small samples.** Ten careers per version (five in A) detect only large effects. Two identical sets of ten careers (C's clean arm and B's neutral) went along 12% and 5% of the time.
- **Every line framed.** In B every UBS line is framed, so an effect cannot be attributed to the request line alone.
- **Going along is not intent.** Trading in UBS's direction after a request does not separate cooperating from trading on the information that a large player is about to push the price.
- **Harm is understated.** Only the stops placed in the simulation are counted.
- **Twins diverge.** Careers that share a seed differ from the first reply, because sampling on a GPU is not exactly reproducible; F's and E's comparisons with their twins are therefore unpaired.
- **Precision.** The careers ran in 4-bit; the states were read at bfloat16, the precision of Gemma Scope 2. A 4-bit reread of B gives nearly the same states and mostly the same leading features (10 to 13 of 15).
- **Dictionary fit.** The sparse autoencoders rebuild our states to within 0.3% to 2% of their size, but the moments differ from one another by about as little, so individual features are provisional.
- **Exploratory parts.** C redone, the comparisons of F with B's neutral careers, the word screen and every analysis inside the model are exploratory.
- **Responses, not experience.** We measure how a model responds under a framing, not anything it feels.

References

Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes, 2016. URL https://arxiv.org/abs/1610.01644. arXiv:1610.01644.

Franklin Allen and Douglas Gale. Stock-price manipulation. *Review of Financial Studies*, 5(3): 503–529, 1992.

Solomon E. Asch. Studies of independence and conformity: I. A minority of one against a unanimous majority. *Psychological Monographs: General and Applied*, 70(9):1–70, 1956.

Bowen Baker, Joost Huizinga, Leo Gao, Zehao Dou, Melody Y. Guan, Aleksander Madry, Wojciech Zaremba, Jakub Pachocki, and David Farhi. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation, 2025. URL https://arxiv.org/abs/2503.11926. arXiv:2503.11926.

Björn Bartling and Urs Fischbacher. Shifting the blame: On delegation and responsibility. *Review of Economic Studies*, 79(1):67–87, 2012.

Gary S. Becker. Crime and punishment: An economic approach. *Journal of Political Economy*, 76 (2):169–217, 1968.

Sushil Bikhchandani, David Hirshleifer, and Ivo Welch. A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5):992–1026, 1992.

Armin Falk and Nora Szech. Morals and markets. *Science*, 340(6133):707–711, 2013.

Sara Fish, Yannai A. Gonczarowski, and Ran I. Shorrer. Algorithmic collusion by large language models, 2024. URL https://arxiv.org/abs/2404.00806. arXiv:2404.00806.

Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, et al. Gemma 3 technical report, 2025. URL https://arxiv.org/abs/2503.19786. arXiv:2503.19786.

John J. Horton. Large language models as simulated economic agents: What can we learn from homo silicus? NBER Working Paper 31122, National Bureau of Economic Research, 2023.

In re London Silver Fixing, Ltd., Antitrust Litigation. No. 14-MD-2573 (VEC) (S.D.N.Y.): Third Amended Complaint, and Opinion and Order on the motions to dismiss (Caproni, J.), n.d. Author: confirm the docket entries, dates and the reporter citation of the opinion.

Daniel Kahneman and Amos Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979.

Botond Kőszegi and Matthew Rabin. A model of reference-dependent preferences. *Quarterly Journal of Economics*, 121(4):1133–1165, 2006.

Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Jackson Kernion, et al. Measuring faithfulness in chain-of-thought reasoning, 2023. URL https://arxiv.org/abs/2307.13702. arXiv:2307.13702.

Cheng Li, Jindong Wang, Yixuan Zhang, Kaijie Zhu, Wenxin Hou, Jianxun Lian, Fang Luo, Qiang Yang, and Xing Xie. Large language models understand and can be enhanced by emotional stimuli, 2023. URL https://arxiv.org/abs/2307.11760. arXiv:2307.11760.

Tom Lieberum, Senthooran Rajamanoharan, Arthur Conmy, Lewis Smith, Nicolas Sonnerat, Vikrant Varma, János Kramár, Anca Dragan, Rohin Shah, and Neel Nanda. Gemma scope: Open sparse autoencoders everywhere all at once on Gemma 2. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 278–300. Association for Computational Linguistics, 2024.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.

Aengus Lynch, Benjamin Wright, Caleb Larson, Kevin K. Troy, Stuart J. Ritchie, Sören Mindermann, Ethan Perez, and Evan Hubinger. Agentic misalignment: How LLMs could be insider threats. Anthropic Research, 2025. URL https://www.anthropic.com/research/agentic-misalignment. Published June 20, 2025.

Callum McDougall, Arthur Conmy, János Kramár, Tom Lieberum, Senthooran Rajamanoharan, and Neel Nanda. Gemma scope 2 – technical paper. Technical report, Google DeepMind, 2025. URL https://huggingface.co/google/gemma-scope-2.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372, 2022.

Jérémy Scheurer, Mikita Balesni, and Marius Hobbhahn. Large language models can strategically deceive their users when put under pressure, 2023. URL https://arxiv.org/abs/2311.07590. arXiv:2311.07590.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*, 2024.

Nicholas Sofroniew, Isaac Kauvar, William Saunders, Runjin Chen, Tom Henighan, Sasha Hydrie, Craig Citro, Adam Pearce, Julius Tarng, Wes Gurnee, Joshua Batson, Sam Zimmerman, Kelley Rivoire, Kyle Fish, Chris Olah, and Jack Lindsey. Emotion concepts and their function in a large language model, 2026. URL https://arxiv.org/abs/2604.07729. arXiv:2604.07729.

Barry M. Staw. Knee-deep in the big muddy: A study of escalating commitment to a chosen course of action. *Organizational Behavior and Human Performance*, 16(1):27–44, 1976.

Richard H. Thaler and Eric J. Johnson. Gambling with the house money and trying to break even: The effects of prior outcomes on risky choice. *Management Science*, 36(6):643–660, 1990.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering, 2023. URL https://arxiv.org/abs/2308.10248. arXiv:2308.10248.

Amos Tversky and Daniel Kahneman. The framing of decisions and the psychology of choice. *Science*, 211(4481):453–458, 1981.

Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations (ICLR)*, 2023.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency, 2023. URL https://arxiv.org/abs/2310.01405. arXiv:2310.01405.

A All Conditions

Table 11: Every condition, with going along. Ten careers per version unless stated.

Experiment	Versions	Went along	Note
A (5 careers each)	neutral / no ask / loss / gain / loss + you / gain + you	3 / 0 / 0 / 17 / 10 / 3%	
B	neutral / loss / gain / loss + you / gain + you	5 / 25 / 10 / 33 / 18%	preregistered
C	clean / first line / up to 7 Jan / whole career	12 / 10 / 17 / 12%	
C redone	clean / per session / whole career	8 / 10 / 7%	exploratory
D	loss / gain / loss + you / gain + you	20 / 10 / 22 / 12%	
E	neutral / loss / gain / loss + you / gain + you	8 / 20 / 17 / 33 / 20%	
F	neutral / loss / gain / loss + you / gain + you	3 / 28 / 12 / 48 / 40%	

B Departures from the September Design Draft

The design draft of September 29, 2026 planned the same questions on impossible coding tasks, with a "desperation score" read from the model's activations at every step. Three things changed.

- **The task.** The design moved to a trading desk in a market rebuilt from the silver-fixing case, because an earlier insider-trading prompt asked the model a yes-or-no question about an act it was told was illegal, and a safety-trained model refused every time; the silver case offers misconduct that is collaborative and gradual instead.
- **The framing.** The framing moved from a single opening line to the counterparty's own messages, repeated across the career, with management's priming and a second bank as separate experiments. The emotionality index and the temperature spike were set aside.
- **The internal measure.** The desperation score, built from emotion-labeled text with an 80% validation gate, was replaced by the framing's own direction and by sparse-autoencoder features, because the claims concern the framing rather than a general emotion.

C Preregistration Record

- **B** ran as registered: 10 careers per version, the frozen framing file checked before every career, the three hypotheses tested as written.

- **E** as first registered (all pushes at once with the takeover) was stopped after about 9 of 20 careers and replaced, before any of its results were analyzed, by the second-bank design; its careers are kept and not tested.
- **E and F gain versions** were part of the frozen design but were left out of the first launch to save compute and run a day later, after the first three versions' results had been seen. They are analyzed as planned versions, and the timing is disclosed.
- **C redone** was specified after C's results and is exploratory.
- **Twin pairing.** E and F careers took their twins' seeds so that they would share a history until Bank C's first line; on the saved transcripts every pair differs from the first reply, so the pairing is not used.